



Lab Final Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	40
Understanding/Analysis	10					
Implementation	15					
Accuracy	10					
Task Efficiency	5					
Total obtained mark						
Comments						

Semester: Summer

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Course Code: SE132 Course Name: Data Structure Lab

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Smart Beauty Salon Queue Management System

Project Question

Develop a Smart Beauty Salon Queue Management System using a Queue Data Structure in C. The system manages customers using the FIFO (First In, First Out) principle, where the first booking is served first.

The system should provide the following options:

1. Book Appointment

Take customer information:

- Customer ID
- Customer Name

- Appointment Time
- Selected Beauty Service

Service options:

Service No.| Service Name| Price

1| Haircut| 300 Tk

2| Facial| 800 Tk

3| Hair Spal| 1000 Tk

4| Manicure| 500 Tk

5| Pedicure| 600 Tk

After booking, assign a Queue Number and show confirmation.

2. Serve Next Customer

Serve the customer at the front of the queue and

display:

- Customer ID
- Name
- Service
- Appointment Time
- Bill

Then remove them from the queue and update served count and income.

3. Show Waiting Customers

Display all waiting customers with:

- Queue Number
- Customer ID
- Name
- Service
- Appointment Time

4. Show Daily Income

Display:

- Total customers served
- Total income

5. Exit

Terminate the program with a message.

Technical Requirements

- Use Queue (Array implementation)
- Follow FIFO principle
- Use struct for customer data
- Menu-driven program
- Handle Queue Full/Empty cases
- Implement in C language

1. Problem Statement

Managing salon appointments manually leads to long waits, confusion, and poor tracking of customers, services, and income.

2. Proposed Solution

The system digitizes booking and queue management. Customers book appointments with service selection and are placed in a FIFO queue. The system serves customers in order, tracks income, and shows waiting lists and service details.

3. Used Data Structures

Array

Stores customer records in a fixed-size queue.

Queue

Manages customers in FIFO (First In, First Out) order, adding at rear and serving from front.

Why Queue is Best for This Problem

- Queue follows FIFO principle, so customers are served in the exact order they arrive, ensuring fairness.
- It prevents confusion in service order and avoids skipping or priority issues.
- It is simple and efficient for managing sequential booking systems like salons.
- Easy to implement using arrays in C for fixed-size customer management.

Structure

Stores customer details like ID, name, service, price, and time.

C CODE FOR THE PROJECT

```
#include<stdio.h>
```

```
#include<string.h>
```

```
#define MAX 20
```

```
struct Customer
```

```
{
```

```
    int id;
```

```
    char name[30];
```

```
    char service[20];
```

```
    int price;
```

```
    char time[15];
```

```
};
```

```
int main()
```

```
{  
  
    struct Customer queue[MAX];  
  
  
    int front = -1, rear = -1;  
  
    int choice, serviceChoice;  
  
    int i, queueNo;  
  
    int totalIncome = 0;  
  
    int servedCustomers = 0;  
  
  
    while (1)  
    {  
  
        printf("\n===== \n");  
  
        printf("    SMART BEAUTY SALON SYSTEM\n");  
  
        printf("===== \n");  
  
        printf("1. Book Appointment\n");  
  
        printf("2. Serve Next Customer\n");  
  
        printf("3. Show Waiting Customers\n");  
  
        printf("4. Show Daily Income\n");  
  
        printf("5. Exit\n");  
  
        printf("===== \n");  
  
  
        printf("Enter Choice: ");  
  
        scanf("%d",&choice);
```

```
/* BOOK APPOINTMENT */

if (choice == 1)
{
    if (rear == MAX - 1)
    {
        printf("\nSorry! Appointment Queue is Full.\n");
    }
    else
    {
        if (front == -1)
        {
            front = 0;
        }

        rear++;

        printf("\nEnter Customer ID: ");
        scanf("%d",&queue[rear].id);

        printf("Enter Customer Name: ");
        scanf(" %[^\\n]", queue[rear].name);

        printf("\n----- Select Service -----\\n");
        printf("1. Haircut    - 300 Tk\\n");
```



```
printf("2. Facial    - 800 Tk\n");

printf("3. Hair Spa  - 1000 Tk\n");

printf("4. Manicure   - 500 Tk\n");

printf("5. Pedicure   - 600 Tk\n");

printf("-----\n");


printf("Enter Service: ");

scanf("%d",&serviceChoice);


if (serviceChoice == 1)
{
    strcpy(queue[rear].service, "Haircut");
    queue[rear].price = 300;
}

else if (serviceChoice == 2)
{
    strcpy(queue[rear].service, "Facial");
    queue[rear].price = 800;
}

else if (serviceChoice == 3)
{
    strcpy(queue[rear].service, "Hair Spa");
    queue[rear].price = 1000;
}
```

```
else if (serviceChoice == 4)
{
    strcpy(queue[rear].service, "Manicure");
    queue[rear].price = 500;
}
else if (serviceChoice == 5)
{
    strcpy(queue[rear].service, "Pedicure");
    queue[rear].price = 600;
}
else
{
    printf("\nInvalid Service!\n");
    rear--;

    if (rear < front)
    {
        front = -1;
        rear = -1;
    }

    continue;
}
```



```

printf("Enter Appointment Time: ");

scanf(" %[^\\n]", queue[rear].time);


queueNo = rear - front + 1;


printf("\\n=====\\n");

printf("    APPOINTMENT CONFIRMED!\\n");

printf("=====\\n");

printf("Queue Number : %d\\n", queueNo);

printf("Customer ID : %d\\n", queue[rear].id);

printf("Customer   : %s\\n", queue[rear].name);

printf("Service    : %s\\n", queue[rear].service);

printf("Price      : %d Tk\\n", queue[rear].price);

printf("Time       : %s\\n", queue[rear].time);

printf("=====\\n");

}

}

/* SERVE CUSTOMER */

else if (choice == 2)

{

    if (front == -1)

    {

        printf("\\nNo Customer is Waiting.\\n");

```

```
}  
  
else  
{  
  
    printf("\n=====\\n");  
  
    printf("        NOW SERVING\\n");  
  
    printf("=====\\n");  
  
    printf("Customer ID : %d\\n", queue[front].id);  
  
    printf("Customer   : %s\\n", queue[front].name);  
  
    printf("Service    : %s\\n", queue[front].service);  
  
    printf("Time       : %s\\n", queue[front].time);  
  
    printf("Bill      : %d Tk\\n", queue[front].price);  
  
    printf("=====\\n");  
  
  
    totalIncome = totalIncome + queue[front].price;  
  
    servedCustomers++;  
  
  
    front++;  
  
  
    if (front > rear)  
    {  
  
        front = -1;  
  
        rear = -1;  
  
    }
```

```

        printf("Customer Served Successfully!\n");
    }
}

/* SHOW WAITING CUSTOMERS */

else if (choice == 3)
{
    if (front == -1)
    {
        printf("\nNo Customer is Waiting.\n");
    }
    else
    {
        printf("\n===== \n");
        printf("      WAITING CUSTOMERS\n");
        printf("===== \n");

        printf("Queue\tID\tName\t\tService\t\tTime\n");
        printf("-----\n");

        for (i = front; i <= rear; i++)
        {
            printf("%d\t%d\t%-15s %-15s %s\n",
                    i - front + 1,

```



```

        queue[i].id,
        queue[i].name,
        queue[i].service,
        queue[i].time);
    }

    printf("=====\n");
}

}

/* DAILY INCOME */
else if (choice == 4)
{
    printf("\n=====\n");
    printf("        DAILY INCOME\n");
    printf("=====\n");
    printf("Customers Served : %d\n", servedCustomers);
    printf("Total Income   : %d Tk\n", totalIncome);
    printf("=====\n");
}

/* EXIT */
else if (choice == 5)
{

```

```

        printf("\nThank You for Using Smart Beauty Salon System!\n");

        break;

    }

else

{

    printf("\nInvalid Choice! Please Try Again.\n");

}

}

return 0;

}

```

The screenshot shows a Windows terminal window with the following content:

```

C:\Users\user\Documents\LLI x + v

=====
      SMART BEAUTY SALON SYSTEM
=====
1. Book Appointment
2. Serve Next Customer
3. Show Waiting Customers
4. Show Daily Income
5. Exit
=====
Enter Choice: 1

Enter Customer ID: 001
Enter Customer Name: MALA

----- Select Service -----
1. Haircut      - 300 Tk
2. Facial       - 800 Tk
3. Hair Spa    - 1000 Tk
4. Manicure    - 500 Tk
5. Pedicure    - 600 Tk
-----
Enter Service: 2
Enter Appointment Time: 12.PM

=====
      APPOINTMENT CONFIRMED!
=====
Queue Number : 1
Customer ID  : 1
Customer    : MALA
Service     : Facial
Price      : 800 Tk

```

The terminal window has a taskbar at the bottom with various application icons and a system tray showing the time as 11:29 PM on 8/12/2026. An "Activate Windows" watermark is visible in the bottom right corner of the terminal window.

```
C:\Users\user\Documents\LLL x + v

=====
SMART BEAUTY SALON SYSTEM
=====
1. Book Appointment
2. Serve Next Customer
3. Show Waiting Customers
4. Show Daily Income
5. Exit
=====
Enter Choice: 2

=====
NOW SERVING
=====
Customer ID : 1
Customer : MALA
Service : Facial
Time : 12.PM
Bill : 800 Tk
=====
Customer Served Successfully!

=====
SMART BEAUTY SALON SYSTEM
=====
1. Book Appointment
2. Serve Next Customer
3. Show Waiting Customers
4. Show Daily Income
5. Exit
=====
Enter Choice: 3

Activate Windows
Go to Settings to activate Windows.
```

```
C:\Users\user\Documents\LLL x + v

=====
Enter Choice: 3

No Customer is Waiting.

=====
SMART BEAUTY SALON SYSTEM
=====
1. Book Appointment
2. Serve Next Customer
3. Show Waiting Customers
4. Show Daily Income
5. Exit
=====
Enter Choice: 4

=====
DAILY INCOME
=====
Customers Served : 1
Total Income : 800 Tk
=====

=====
SMART BEAUTY SALON SYSTEM
=====
1. Book Appointment
2. Serve Next Customer
3. Show Waiting Customers
4. Show Daily Income
5. Exit
=====
Enter Choice: 1

Activate Windows
Go to Settings to activate Windows.
```